

Q1 Distinguish between Internet, Intranet & Extranet. Structure of Internet, Modes of Connecting, How a Modem Works.

Introduction : The Internet, Intranet, and Extranet are three related but distinct network concepts.

Understanding their differences is fundamental to grasping modern digital communication.

•**Internet:** A global, public network of interconnected computers using TCP/IP protocols. Open to everyone — billions of users, devices, and servers worldwide share information freely.

•**Intranet:** A private network within an organization. Uses the same technologies as the Internet (HTML, HTTP, TCP/IP) but access is restricted to authorized employees. Example: A company’s internal HR portal.

•**Extranet:** An extension of an intranet that selectively allows external parties (vendors, partners, customers) to access specific internal resources. It bridges the private intranet and the public internet.

Key Differences Table

Internet: Public, global, no ownership. Intranet: Private, organizational, owned by company. Extranet: Semi-private, shared with trusted partners.

Structure of the Internet

The Internet is structured as a layered, hierarchical network:

•Tier 1 Networks: Large backbone providers (AT&T, NTT) interconnect globally via peering agreements. These form the "backbone" of the Internet.

•Tier 2 & 3 Networks: Regional and local ISPs connect to Tier 1 networks and provide Internet access to end users and businesses.

•End Devices: PCs, smartphones, servers — all connected via ISPs.

•Data travels as packets, routed across multiple networks using routers, switches, and fiber-optic or wireless links.

How the Internet Works

When you visit a website, your browser sends an HTTP request. Your ISP routes this request through multiple routers. The destination server receives the request and sends back data in packets. Packets travel independently, may take different routes, and are reassembled at the destination. This entire process relies on the TCP/IP protocol suite.

Modes of Connecting to the Internet

•Dial-Up: Uses telephone lines and a modem. Slow (max 56 kbps). Nearly obsolete today.

•DSL (Digital Subscriber Line): Uses existing telephone wiring but provides much faster speeds. ADSL (Asymmetric DSL) is most common — faster download than upload.

•Cable: Uses TV cable infrastructure. Faster than DSL. Speeds up to hundreds of Mbps.

•Fiber Optic: Uses light signals in glass/plastic fiber. Extremely high speed (Gbps range). Most modern broadband technology.

•Satellite: Useful in remote areas. Higher latency. Providers like Starlink offer low-earth-orbit satellite internet.

•Mobile (3G/4G/5G): Cellular network-based internet access. 5G offers ultra-low latency and multi-Gbps speeds.

•Wi-Fi: Wireless local area connection to a router which itself connects to the Internet via one of the above methods.

•Leased Lines: Dedicated private connections for businesses. Guaranteed bandwidth and uptime.

How a Modem Works

A modem (MOdulator-DEModulator) is a hardware device that converts digital data from a computer into analog signals suitable for transmission over telephone lines, and vice versa.

Modulation (Sending Data)

When you send data, the modem takes the binary digital data (0s and 1s) from your computer and modulates (converts) it into an analog signal — varying the frequency, amplitude, or phase of a carrier wave — so it can travel over copper telephone wires.

Demodulation (Receiving Data)

When receiving, the modem does the reverse: it demodulates the incoming analog signal, extracting the encoded digital data and passing it to the computer.

Q2 IP Addressing Scheme — Classes A, B, C, D, E. DNS, ISP, Internet Congestion, Collaborative Computing.

Introduction to IP Addressing

An IP (Internet Protocol) address is a unique numerical label assigned to every device connected to a network. It enables devices to identify and communicate with each other on the Internet or on a local network.

IP Addressing Scheme

IPv4 addresses are 32-bit numbers, written in dotted-decimal notation as four octets separated by dots.

Example: 192.168.1.10.

Each octet can range from 0 to 255, giving a theoretical maximum of over 4 billion unique addresses. The address is divided into a Network part (identifies the network) and a Host part (identifies the device within that network).

How Addresses Are Assigned

•Static Assignment: A fixed IP manually configured by an administrator. Used for servers and network devices.

•Dynamic Assignment: The DHCP (Dynamic Host Configuration Protocol) server automatically assigns an IP from a pool when a device connects. Most home and office devices use this.

•IANA (Internet Assigned Numbers Authority) oversees global allocation; IANA assigns to Regional Internet Registries (RIRs), which assign to ISPs, who assign to end users.

IP Address Classes

IPv4 addresses are grouped into 5 classes based on the leading bits of the address:

•Class A: Range 1.0.0.0 – 126.255.255.255. First bit: 0. Default mask: 255.0.0.0 (/8). Supports ~16 million hosts per network. Used for very large organizations.

•Class B: Range 128.0.0.0 – 191.255.255.255. First bits: 10. Default mask: 255.255.0.0 (/16). Supports ~65,000 hosts per network. Used for medium organizations.

•Class C: Range 192.0.0.0 – 223.255.255.255. First bits: 110. Default mask: 255.255.255.0 (/24). Supports 254 hosts per network. Most commonly used.

•Class D: Range 224.0.0.0 – 239.255.255.255. Reserved for multicast (sending data to multiple recipients simultaneously).

•Class E: Range 240.0.0.0 – 255.255.255.255. Reserved for experimental use. Not available for general assignment.

DNS (Domain Name System)

DNS is often called the "phonebook of the Internet." It translates human-readable domain names (like www.google.com) into machine-readable IP addresses (like 142.250.80.36).

•DNS Resolver: Your ISP’s server that accepts DNS queries from clients.

•Root Servers: 13 sets of root name servers hold the top-level domain information.

•TLD Servers: Handle top-level domains like .com, .org

•Authoritative Name Servers: Hold the actual DNS records for specific domains.

ISP (Internet Service Provider)

An ISP is a company that provides access to the Internet. They maintain the infrastructure (servers, routers, cables) and provide connectivity to homes and businesses in exchange for a subscription fee. ISPs operate at Tier 1, 2, or 3 levels.

Collaborative Computing

Collaborative computing refers to the use of networked computers and software to enable multiple users to work together on shared tasks — e.g., Google Docs, Microsoft 365, video conferencing tools.

Internet Congestion

Internet congestion occurs when the volume of data traffic on a network segment exceeds its capacity, causing slowdowns, packet loss, and increased latency.

•Causes: High number of simultaneous users, bandwidth-intensive applications (video streaming), insufficient infrastructure capacity, network attacks (DDoS).

•Effects: Slow page loads, video buffering, dropped packets, VoIP call degradation.

•Solutions: Traffic shaping and QoS (Quality of Service) prioritizes critical traffic. CDNs (Content Delivery Networks) like Cloudflare distribute content geographically. Load balancing distributes traffic across multiple servers. Network upgrades (fiber optic, 5G) increase capacity.

Q3 Explain FTP, HTTP, Telnet & Gopher. Role of HTTP & Web Browsers in WWW. Search Engines: Directory vs Meta. Introduction

The World Wide Web relies on several application-layer protocols for communication between clients and servers. Understanding FTP, HTTP, Telnet, and Gopher helps explain the evolution of network services.

FTP (File Transfer Protocol)

FTP is a standard protocol for transferring files between a client and server over a TCP/IP network. It uses two channels: port 21 for commands and port 20 for data transfer.

•Active FTP: Server initiates the data connection back to the client.

•Passive FTP: Client initiates both connections (firewall-friendly).

•FTP transfers files in binary or ASCII mode. SFTP and FTPS are secure variants.

Telnet

Telnet is a protocol that provides a command-line interface to a remote computer over a network (port 23). It allows users to log in and execute commands on remote machines. •Major limitation: Telnet transmits data in plain text — including passwords — making it insecure. Replaced by SSH (Secure Shell) which encrypts all communication.

Gopher

Gopher (port 70) was an early Internet protocol for distributing, searching, and retrieving documents. It presented a hierarchical menu-based system (like a directory tree) for navigating content.

•Predated the World Wide Web. Uses a simple text-based menu structure.

•Largely obsolete; replaced by HTTP/WWW which supports richer content including multimedia.

HTTP & Web Browsers in the WWW

HTTP (HyperText Transfer Protocol) is the foundation of data communication on the Web. It operates as a request-response protocol:

•Client (browser) sends an HTTP Request to the server (e.g., GET /index.html HTTP/1.1).

•Server processes the request and sends back an HTTP Response with status code (200 OK, 404 Not Found, etc.) and the requested content.

•HTTP is stateless — each request is treated independently with no memory of prior interactions.

•HTTPS (HTTP Secure) encrypts communication using TLS/SSL, protecting data in transit.

Web browsers parse and render HTML, CSS, and JavaScript, displaying rich, multimedia web pages. Browsers also handle cookies, caching, and browser history.

Popular Web Browsers

•Google Chrome: Most widely used. Fast, secure, extensive extension library.

•Mozilla Firefox: Open-source, privacy-focused.

•Microsoft Edge: Built on Chromium, integrated with Windows.

•Apple Safari: Default browser on macOS and iOS. Optimized for Apple hardware.

•Opera: Known for built-in VPN and data-saving features.

•Brave: Privacy-first browser with built-in ad blocker.

How a Search Engine Works

A search engine has three main components:

•1. Crawling: Automated bots (spiders/crawlers) browse the web by following hyperlinks from page to page, downloading content for indexing.

•2. Indexing: The downloaded content is analyzed and stored in a massive database (the index). Pages are categorized by keywords, metadata, and relevance signals.

•3. Ranking/Retrieval: When a user enters a query, the engine searches its index and applies ranking algorithms (like Google PageRank) to return the most relevant results in order of relevance.

Directory vs Meta Search Engines

•Directory Search Engines: Maintain a human-curated hierarchy of websites organized into categories. Example: Yahoo! Directory (now discontinued), DMOZ. Pros: High quality, organized. Cons: Limited coverage, slow to update.

•Meta Search Engines: Do not maintain their own index. Instead, they send queries to multiple search engines simultaneously and aggregate/compile the results.

Examples: Dogpile, MetaCrawler. Pros: Broader coverage. Cons: Less control over result quality, slower response.

•Standard Search Engines (Web Crawlers): Maintain their own index via crawling. Examples: Google, Bing, DuckDuckGo. Most powerful and widely used.

Q4 TCP/IP Model & Core Protocols. What is HTML? Basic Structure, Important Tags, Hyperlinks & Plug-in Architecture. Introduction to TCP/IP Model

The TCP/IP (Transmission Control Protocol / Internet Protocol) model is the practical framework that governs how data is transmitted across the Internet. It consists of four layers (a simplified alternative to the 7-layer OSI model).

TCP/IP Layers & Core Protocols

- 1. Application Layer: The topmost layer, directly interacting with software applications. Protocols: HTTP/HTTPS (web), FTP (file transfer), SMTP/POP3/IMAP (email), DNS (name resolution), SSH (secure shell), Telnet.
- 2. Transport Layer: Provides end-to-end communication between applications. TCP (Transmission Control Protocol): Connection-oriented, reliable, ordered delivery with error checking. Three-way handshake (SYN, SYN-ACK, ACK). UDP (User Datagram Protocol): Connectionless, fast, no guaranteed delivery. Used for streaming, gaming.
- 3. Internet Layer: Responsible for logical addressing and routing packets across networks. Core protocol: IP (Internet Protocol). Also includes ICMP (ping/traceroute) and ARP (address resolution).
- 4. Network Access Layer (Link Layer): Handles physical transmission of data over the local network. Protocols: Ethernet, Wi-Fi (802.11), MAC addressing.

HTML — Introduction

HTML (HyperText Markup Language) is the standard markup language used to create web pages. It describes the structure and content of a web page using a series of elements represented by tags.

- HTML is not a programming language — it cannot perform computations. It describes document structure.
- An HTML document is a plain text file with special markup tags. Browsers read HTML and render the visual web page.
- Current standard: HTML5.

Basic HTML Structure

Every HTML document follows this skeleton:
<!DOCTYPE html> — declares the document type.
<html> — root element.
<head> — metadata (title, CSS links, scripts).
<title> — tab/window title.
<body> — visible content of the page.

Important HTML Tags

- <h1> to <h6>: Headings, h1 being the largest.
- <p>: Paragraph text.
- : Hyperlink (anchor tag).
- : Embed an image.
- , , : Unordered / ordered lists.
- <table>, <tr>, <td>, <th>: Tables.
- <form>, <input>, <button>: Forms for user input.
- <div>, : Generic container elements for layout/styling.

How Hyperlinks Are Created

- Hyperlinks are created using the anchor (<a>) tag with the href (Hypertext REference) attribute:
- Visit Google
- External links: Full URL (https://...)
 - Internal links: Relative path (/about.html)
 - Email links: href="mailto:user@example.com"
 - Phone links: href="tel:+91XXXXXXXXXX"
 - Anchor links: href="#section-id" (jumps to a section on the same page)
 - target="_blank" opens the link in a new tab.

Architecture of Plug-ins

- Plug-ins (also called plugins or browser plugins) are software components that extend the functionality of a web browser beyond what HTML/CSS/JS natively provide.
- Integration: Plug-ins are integrated into the browser framework. When the browser encounters content that requires a plugin (identified by the MIME type), it loads the plugin to handle that content inline within the web page.
 - Examples: Adobe Flash Player (for animations/games), Adobe Reader (PDFs in browser), Java Applet plugin, QuickTime (media).

- Plugin API: Browsers expose a Netscape Plugin API (NPAPI) or newer PPAPI (Pepper Plugin API in Chrome) that allows plugins to interface with browser rendering.

Plug-in Architecture Details

- Discovery: On page load, browser checks MIME types of embedded content (e.g., application/pdf). If a registered plugin handles that MIME type, the plugin is invoked.
- Rendering: The plugin renders its content in a rectangular area within the browser window. The browser manages the surrounding HTML; the plugin manages its own area.
- Communication: Plugins can communicate with JavaScript via the DOM (Document Object Model), enabling dynamic interaction.
- Security Concerns: Plugins (especially Flash) became major security vulnerabilities. Most modern browsers have deprecated NPAPI plugins

Q5 Components of E-mail Architecture. How E-mail Works. SMTP & MIME — Header Fields, Types, Mailer Features & Message Composition.

Introduction

Electronic mail (e-mail) is one of the most widely used services on the Internet. Understanding e-mail architecture, protocols, and the MIME extension is essential for any IT professional.

Components of E-mail Architecture

- MUA (Mail User Agent): The client application used to compose, read, send, and receive emails. Examples: Outlook, Thunderbird, Gmail web interface.
- MTA (Mail Transfer Agent): The server software that transfers email between mail servers using SMTP. Examples: Sendmail, Postfix, Microsoft Exchange.
- MDA (Mail Delivery Agent): Receives email from the MTA and stores it in the user's mailbox. Also called Local Delivery Agent (LDA).
- Mail Server: The computer running the MTA/MDA software. Stores user mailboxes and handles routing.
- DNS (Mail Exchanger records): MX records in DNS tell sending MTAs which server accepts email for a domain (e.g., @gmail.com → Google's mail servers).
- Mailbox: The storage location for a user's received email, held on the mail server or downloaded to the client.

How an E-mail System Works

When you send an email, the following sequence of events occurs:

- 1. You compose and send the email in your MUA (e.g., Gmail). The MUA connects to your outgoing mail server (SMTP server) on port 25, 465, or 587.
- 2. Your SMTP server looks up the recipient domain's MX record in DNS to find the destination mail server.
- 3. Your MTA connects to the recipient's MTA and transfers the email via SMTP.
- 4. The recipient's MTA passes the email to the MDA, which stores it in the user's mailbox.
- 5. When the recipient checks their email, their MUA connects to the incoming server using POP3 (Post Office Protocol) or IMAP (Internet Message Access Protocol) to retrieve the message.

SMTP (Simple Mail Transfer Protocol)

SMTP is the protocol used for sending email between mail servers (MTA to MTA) and from client to server. It operates over TCP port 25 (server-to-server), port 587 (client submission), or port 465 (SMTPS — secure). SMTP is a push protocol — the sender pushes email to the receiving server.

MIME — What It Is

MIME (Multipurpose Internet Mail Extensions) is an extension to the original SMTP email protocol that allows email to carry content beyond plain ASCII text — including attachments, images, audio, video, and HTML.

- Without MIME, email could only carry plain ASCII text.

MIME introduced the ability to attach any type of file and include multipart messages.

MIME Header Fields

- MIME-Version: Specifies the MIME version (typically "1.0").
- Content-Type: Specifies the media type and subtype of the content. E.g., text/html, image/jpeg, application/pdf, multipart/mixed.
- Content-Transfer-Encoding: How the content is encoded for transmission. Values: 7bit, 8bit, base64 (for binary data), quoted-printable (for mostly ASCII with some special chars).
- Content-Disposition: Whether content is inline or an attachment. E.g., attachment; filename="doc.pdf".
- Content-ID: Unique identifier for a MIME part.
- Content-Description: Human-readable description of the content part.

MIME Types

- Text: text/plain, text/html, text/css
- Image: image/jpeg, image/png, image/gif, image/svg+xml
- Audio: audio/mpeg (MP3), audio/wav, audio/ogg
- Video: video/mp4, video/ogg, video/webm
- Application: application/pdf, application/zip, application/json, application/octet-stream (generic binary)
- Multipart: multipart/mixed (email with attachments), multipart/form-data (web form uploads), multipart/alternative (email with both plain text and HTML versions)

Mailer Features & Message Components

- Message Components: Header (From, To, CC, BCC, Subject, Date, Message-ID) and Body (the actual message content, possibly multipart).
- Message Composition: MUAs allow rich-text editing, attachments, signatures, and templates.
- Mailer Features: Address book/contacts, folder organization, filters/rules, search, spam filtering, calendar integration (e.g., meeting invitations), encryption (S/MIME, PGP).

Q6 HTML vs DHTML vs XML. JavaScript — Overview, Advantages & Dev Tools. Server-Side Programming. Notes on Chat Rooms, Newsgroups & Mailing Lists.

Introduction

The World Wide Web has evolved through several markup and scripting technologies — from basic HTML to dynamic DHTML and structured XML — complemented by powerful server-side and client-side scripting.

HTML vs DHTML vs XML

- HTML (HyperText Markup Language): A static markup language for creating web page structure and content. Fixed layout; requires page reload. All browsers can render HTML.
- DHTML (Dynamic HTML): Not a separate language but a combination of HTML + CSS + JavaScript + DOM. Allows web pages to change dynamically without reloading. Elements can appear/disappear, animate, or respond to user events in real time. Example: dropdown menus, image sliders.
- XML (eXtensible Markup Language): A meta-language for defining custom markup languages. Designed to store and transport data, not display it. Tags are user-defined and self-descriptive. Platform and language neutral. Example: RSS feeds, configuration files.

JavaScript as a Scripting Language

JavaScript is the dominant client-side scripting language of the Web.

- Client-side: Runs in the browser, no server interaction required for basic operations. enables real-time interactivity.
- Event-driven: Responds to user actions through event handlers like onclick, onmouseover, onsubmit.
- Interpreted: No compilation needed. The browser's JavaScript engine (V8 in Chrome) executes the code.
- Object-based (not fully OOP): Supports objects and prototype-based inheritance but does not have classical class-based OOP (though ES6+ added class syntax).
- Case-sensitive: myVar and myvar are different identifiers.

Advantages of JavaScript

- Reduces server load by validating forms client-side before submission.
- Creates rich, interactive user interfaces
- Asynchronous programming via AJAX allows updating parts of a page without full reload.
- Massive ecosystem: Node.js enables server-side JavaScript; frameworks like React, Angular, Vue.

JavaScript Development Tools

- Browser DevTools: Chrome DevTools, Firefox Developer Tools — debugging, profiling, DOM inspection, network monitoring. Available by pressing F12.
- Text Editors/IDEs: VS Code, WebStorm, Sublime Text — offer syntax highlighting, auto-complete, and debugging.
- Package Managers: npm (Node Package Manager), yarn — manage JS libraries and dependencies.
- Build Tools: Webpack, Babel — bundle modules and transpile modern JS to older compatible versions.
- Testing Frameworks: Jest, Mocha — automated unit testing.

Server-Side Programming Concepts

Server-side programming refers to scripts that run on the web server in response to client requests. Unlike client-side code which runs in the browser, server-side code runs on the server and generates dynamic HTML that is then sent to the client.

- Technologies: PHP, Node.js, Python (Django/Flask), Java (Servlets/JSP), Ruby on Rails, ASP.NET.
- Tasks: Database queries, authentication/authorization, file I/O, business logic processing, session management, sending emails.
- The client (browser) never sees the server-side source code — only the output HTML is sent.

Chat Rooms

A chat room is an online platform where multiple users can send and receive text messages in real time. Messages appear to all participants simultaneously. Modern chat rooms use WebSockets for real-time bidirectional communication (e.g., Discord, Slack). Early chat rooms used IRC (Internet Relay Chat).

Newsgroups

Newsgroups (Usenet groups) are an early Internet discussion system, predating the Web. They function like bulletin boards where users post messages organized by topic (e.g., comp.lang.java). Messages are distributed across Usenet servers globally using the NNTP protocol. Usenet groups were categorized into hierarchies: comp (computers), sci (science), rec (recreation), soc (society), etc.

Mailing Lists

A mailing list is a collection of email addresses grouped under a single address. When a message is sent to the list address, it is forwarded to all subscribers. Used for announcements (one-way) or discussions (two-way, where members can reply). Software: Mailman, Listserv.

Q7 What is a Web Server? Differentiate Apache, IIS & Microsoft PWS. Apache Configuration Directives & How to Start/Stop Servers.

Introduction

A web server is a software application (and/or hardware) that stores, processes, and serves web pages to clients (browsers) over the HTTP/HTTPS protocol. When a user requests a URL, the web server locates the resource and sends it back as an HTTP response.

- Core functions: Accepting HTTP requests, locating the requested resource (file, script output), returning HTTP responses, managing virtual hosts, logging.
- Popular web servers: Apache HTTP Server, Microsoft IIS, Nginx, LiteSpeed, Microsoft PWS.

Apache HTTP Server

Apache is the most widely deployed open-source web server, developed by the Apache Software Foundation. It runs on Linux, Windows, macOS, and most Unix-like systems. Apache uses a modular architecture — functionality is added via modules (e.g., mod_ssl for HTTPS, mod_rewrite for URL rewriting, mod_php for PHP support).

- Default configuration file: httpd.conf
- Apache uses a process/thread-based model, creating a new process or thread for each connection.

IIS (Internet Information Server)

IIS (Internet Information Services) is Microsoft’s web server, included with Windows Server. It is tightly integrated with the Windows OS and supports ASP.NET, PHP, and classic ASP.

- Configured via the IIS Manager GUI or XML-based configuration files (web.config).

- Supports Windows Authentication, integrated with Active Directory.
- Uses ISAPI (Internet Server Application Programming Interface) for extensions.
- Best suited for Windows-based hosting environments.

Microsoft PWS (Personal Web Server)

PWS was a lightweight version of IIS designed for personal/development use on Windows 95/98/NT Workstation. It allowed developers to test web pages locally without a full server setup.

- Supported basic HTTP, ASP (Active Server Pages), and FrontPage extensions.

- Not suitable for production; replaced by IIS for development on modern Windows with IIS Express.

Apache vs IIS vs PWS — Key Differences

- Apache: Open-source, cross-platform, module-based, highly configurable, large community.
- IIS: Microsoft-proprietary, Windows-only, deeply integrated with Windows/.NET, strong GUI management.
- PWS: Lightweight, for personal/dev use only, Windows 9x era, now obsolete.

Apache Configuration Directives

Apache is configured primarily through the httpd.conf file (and supplementary .conf files in the conf.d/ directory). Key directives include:

- ServerRoot: Base directory for server files. E.g., ServerRoot "/etc/httpd"
- Listen: Port(s) to listen on. E.g., Listen 80 or Listen 443
- DocumentRoot: The directory from which Apache serves files. E.g., DocumentRoot "/var/www/html"
- DirectoryIndex: Default file to serve for a directory request. E.g., DirectoryIndex index.html index.php
- ServerName: The server’s hostname and port. E.g., ServerName www.example.com:80
- <Directory>: Sets permissions and options for a directory. Options Indexes FollowSymLinks; AllowOverride All; Require all granted.

- VirtualHost: Configures virtual hosting to serve multiple websites from one server.
- ErrorLog / CustomLog: Define log file locations and formats.
- LoadModule: Loads additional Apache modules.
- <Location>: Applies directives to a URL path rather than a filesystem path.

Starting and Stopping Apache

On Linux/Unix systems, Apache (httpd or apache2) is managed via system commands:

- Start: sudo systemctl start apache2 (Ubuntu/Debian) OR sudo service httpd start (RHEL/CentOS)
- Stop: sudo systemctl stop apache2
- Restart: sudo systemctl restart apache2 (full restart) OR sudo systemctl reload apache2 (graceful reload without dropping connections)
- Status: sudo systemctl status apache2 — shows if running, uptime, recent logs.
- Enable on boot: sudo systemctl enable apache2.

On Windows, Apache can be started/stopped via the Apache Service Monitor in the system tray, via the Services console (services.msc), or via command line: net start Apache2.4 / net stop Apache2.4.

Q8 Threats & Attacks to Internet Security. Symmetric & Asymmetric Encryption. Digital Signatures. Firewalls (Types). IDS & Why IDS is Preferred Over Firewalls.

Introduction to Internet Security

As reliance on the Internet grows, so do threats to its security. Understanding these threats and the mechanisms used to counter them is critical for protecting data, systems, and users.

Threats & Attacks to Internet/Computer Security

- Malware: Malicious software — viruses, worms, Trojans, ransomware, spyware. Spreads via email attachments, malicious websites, infected drives.
- Phishing: Deceptive emails/websites trick users into revealing sensitive information (passwords, credit card numbers) by impersonating legitimate entities.
- DoS / DDoS: Denial-of-Service / Distributed DoS. Floods a server with so much traffic it becomes unavailable to legitimate users. DDoS uses a botnet.
- Man-in-the-Middle (MITM): Attacker secretly intercepts and possibly alters communication between two parties. Common on unsecured Wi-Fi networks.
- SQL Injection: Inserting malicious SQL code into web form inputs to manipulate or dump a database.
- Cross-Site Scripting (XSS): Injecting malicious scripts into web pages viewed by other users.
- Brute Force: Systematically trying all possible passwords until the correct one is found.
- Zero-Day Exploit: Attack targeting an unknown (unpatched) software vulnerability.

Encryption Schemes

- Symmetric Encryption: Same key is used for both encryption and decryption. Fast and efficient for large amounts of data. Key distribution problem: how to securely share the key. Examples: AES (Advanced Encryption Standard), DES (Data Encryption Standard), 3DES. Use case: Encrypting files, database encryption, disk encryption.
- Asymmetric Encryption (Public Key Cryptography): Uses a key pair — a public key (shared with everyone) and a private key (kept secret). Data encrypted with the public key can only be decrypted with the private key. Slower than symmetric but solves the key distribution problem. Examples: RSA, ECC (Elliptic Curve Cryptography), Diffie-Hellman. Use case: SSL/TLS (HTTPS), email encryption (PGP).
- Hybrid Encryption: Most real-world systems (like TLS/HTTPS) use both — asymmetric encryption to securely exchange a symmetric session key, then symmetric encryption for the actual data transfer. This combines speed and security.

Digital Signatures

A digital signature is a cryptographic mechanism that proves the authenticity and integrity of a digital document.

- How it works: The sender hashes the message (creates a fixed-length digest), then encrypts the hash with their private key. This encrypted hash is the digital signature. The recipient decrypts the signature with the sender’s public key to retrieve the hash, then independently hashes the received message. If the two hashes match, the signature is valid — confirming the message came from the stated sender and has not been tampered with.
- Provides: Authentication (proves sender identity), Integrity (message unchanged), Non-repudiation (sender cannot deny sending).

Firewalls — Types

- Packet Filtering Firewall: Inspects each packet’s header (source/destination IP, port, protocol) against rules. Fast but does not inspect packet payload. Examples: iptables
- Stateful Inspection Firewall: Tracks the state of active connections. More intelligent — knows whether an incoming packet is part of an established connection
- Application/Proxy Firewall: Operates at the application layer. Acts as an intermediary (proxy) between client and server, deeply inspecting HTTP, FTP, DNS traffic.
- Next-Generation Firewall (NGFW): Combines stateful inspection, deep packet inspection (DPI), intrusion prevention, SSL decryption, and identity awareness. Examples: Palo Alto, Fortinet, Cisco Firepower.

IDS (Intrusion Detection System)

An IDS monitors network traffic or host activity for suspicious patterns and generates alerts when potential intrusions or policy violations are detected.

- NIDS (Network-based IDS): Monitors the entire network by analyzing captured packets at strategic points.
- HIDS (Host-based IDS): Monitors a single host — system calls, file integrity, log analysis. Example: OSSEC, Tripwire.
- Signature-based detection: Compares traffic against a database of known attack signatures. Fast and accurate for known threats but misses new/unknown attacks.
- Anomaly-based detection: Establishes a baseline of normal behavior and flags deviations. Can detect new attacks but may have higher false-positive rates.

Why IDS Is Preferred Over Firewalls

Firewalls are perimeter-based — they block or allow traffic based on rules but cannot detect sophisticated attacks that pass through legitimate channels. Reasons IDS is preferred/complementary:

- Insider threats: Firewalls protect only external traffic. IDS detects malicious activity from inside the network (disgruntled employees, compromised internal machines).
- Encrypted traffic: Firewalls may pass encrypted traffic without inspection. IDS (especially with SSL decryption) can analyze content for malicious patterns.
- Post-breach detection: If a firewall is bypassed, IDS can still detect and alert on malicious activity inside the network.
- Forensics: IDS logs detailed attack information useful for forensic analysis and understanding breach scope.